OSF Amendment 3 – Correction of the Phase 2 Confirmatory Family (H5)

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0. Registration status

This amendment corrects two of the three confirmatory tests registered in Amendment 2 (https://osf.io/zd8jx, filed 11 October 2026). It is filed before any SHRP2 data have been accessed, in any version. The original registration (https://osf.io/y5zrp, 4 October 2026), its nine tests of H1–H4, Amendment 1 (https://osf.io/cfdkj) and the Phase 2 method are unchanged.

1. Reason for the amendment

The analysis code for H5a–H5c was written and tested on synthetic data with known answers before any SHRP2 data exist. The tests showed that two of the three registered tests, as worded, cannot fail, whatever the data.

- **H5c.** Under the registered leave-one-test-unit-out design, each per-unit threshold t_j lies on the fixed grid, and unit u exceeds ε exactly when its threshold from the other units, λ_u, is smaller than t_u. Because λ_u is the r-th smallest of the other $n - 1$ values of t, with $r = \lceil n(1 - \delta) \rceil$, at most $n - r$ units can exceed ε (exactly $n - r$ without ties). The count x is therefore at most $n - r$, and for every $n \geq 50$ the registered p-value is at most $P(\mathrm{Bin}(n, 0.20) \leq n - r) \leq 0.048$. The outcome is decided by n alone. Simulations confirmed this for $n = 20$, 45, 50 and 100 at all three levels.
- **H5a.** For the same reason, the HP share of units above 5% is at most $(n - r)/n \approx 0.10$ by construction. The pooled split-conformal threshold (C1) controls the average false-alarm rate, so about half of the units lie above 5% even when all units are identical (0.42–0.62 in simulations, unit sizes 150 to 40,000 frames). The registered difference is therefore positive whatever the data.
- **H5b** depends on the data: in the same simulations the recall difference crossed the margin of −0.15 depending on how much the units differ. It is unchanged.

2. Corrected hypotheses

H5a and H5c are re-specified on the deployment design that Amendment 2 already defines as a secondary analysis: the HP threshold is set once on the k eligible validation units and applied to every one of the n eligible test units, separately for each training

seed. Test units never enter the threshold, so the count of test units above ε is not fixed by ranks. Because the validation units also took part in model selection, the test asks a real question: does per-unit control carry over from the units used to build and tune the model to new units?

Test	Claim	Design and levels	Statistic and test	Supported if
H5a	Per-unit control carries over to new units: fewer than 20% of test units exceed the 5% target	Deployment design; HP with ε = 0.05, δ = 0.10, threshold from the k validation units	x = number of test units with FA > 0.05, averaged over seeds and rounded up; exact one-sided binomial test of H0: share ⩾ 0.20, p = P(Bin(n, 0.20) ⩽ x)	Holm-adjusted p < 0.05
H5b	Unchanged from Amendment 2: HP′s recall at 3 s is non-inferior to C1′s, margin 0.15	Leave one test unit out, ε = α = 0.05	recall(HP) − recall(C1); paired cluster bootstrap, $\delta_0 = -0.15$	Holm-adjusted p < 0.05
H5c	Operational validity carries over: fewer than 20% of test units exceed one alarm frame per hour	Deployment design; HP with ε_op = 1/36,000 per frame, δ = 0.10	as H5a, with FA > ε_op	Holm-adjusted p < 0.05

- **Why the binomial test is valid here.** Given the deployed threshold, the test units are independent draws, so the count of test units above ε is binomial with the true share of the deployed threshold. Averaging the count over seeds and rounding up is conservative. In simulations, the test rejected in at most 0.3% of replicates when the true share was 0.21, rejected in 66% (n = 50) and 95% (n = 100) of replicates with 100 validation units when the units were exchangeable, and never rejected when the validation units looked safer than new units (true shares 0.32 and 0.63).
- **Family.** H5a, H5b and H5c remain one family with Holm control at 0.05. Across this family and the registered nine-test family, the family-wise error stays at most 0.10. Holm over all 12 tests is reported as a sensitivity analysis.
- **Not testable.** H5a and H5c are not testable if fewer than 9 validation units or fewer than 50 test units are eligible. The rules for H5b are unchanged (fewer than 20 test units, or fewer than 50 test events).

- **Margins.** The null share of 0.20 is the tolerance already registered for H5c in Amendment 2; it is now used for both H5a and H5c. The practical-relevance label that Amendment 2 attached to H5a no longer applies.

3. Pre-specified descriptive analyses

Reported in full, without p-values:

1. The original H5a and H5c statistics on the leave-one-out design: the share of test units above 5% under HP and under C1, with bootstrap intervals, and the share above one alarm frame per hour. When reported, it is stated that the HP shares are at most (n − r)/n by construction and that C1 controls only the average.
2. For HP and C1 in both designs: the median, 90th percentile and maximum of the per-unit false-alarm rate, and the mean and pooled false-alarm rate of C1 against α. A departure from exchangeability shows in these quantities.
3. Alarm frames per hour and recall at every level, for both designs.

The secondary analyses of Amendment 2 remain: H5a and H5b at the 10% level (H5a in its corrected form), the expectation form (CRC), the published comparators, the strata, and the registered sensitivity analyses.

4. Correction to the development evidence in Amendment 2, Section 7

Under leave-one-out designs the HP shares reported there follow from the same rank identity: 4 of 45 recordings (E1a) and 1 of 12 locations (E2, location level) equal n − r. The pooled method's shares of 36–58% of units above target are close to what identical units give, so they are not evidence of heterogeneity. The development evidence that thresholds break under shift lies in the mean or pooled false-alarm rate: 4 of 12 unseen locations up to 8.0% at a 5% target (Phase 1), pooled rates of 5.3% at 5% and 10.4% at 10% at new locations, and 26% at 5% when INTERACTION thresholds were carried to highD. The random-partition design (E1b, where the held-out unit is separate from the calibration units: 7.9–8.3%) and the transfer of recording-level thresholds to new locations (0 of 12 above target) remain valid checks.

5. What stays unchanged

Everything else in Amendment 2: the method, score, negatives, units, eligibility, grid, levels, H5b, the bootstrap, the secondary and exploratory analyses, and the contingency rules.

6. Process note

The issue was found on 11 October 2026 while unit-testing the analysis script on synthetic data: a Monte Carlo check returned an HP share with almost no spread, which led to the rank argument above. No SHRP2 data were involved. The analysis script, its tests and this amendment are committed to the project repository before any SHRP2 access.

Use of AI assistance. As stated in the original registration, code development and drafting were assisted by an AI system; all analyses are run, checked and approved by the author.